

bookit_lang_tool.py

Commands:

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python3 bookit_lang_tool.py unused [FILE] [--json out.json]
python3 bookit_lang_tool.py report [FILE] [--json out.json]
python3 bookit_lang_tool.py sort [FILE] [--check]
python3 bookit_lang_tool.py prune [FILE] [--force]
python3 bookit_lang_tool.py sync [--en F] [--de F] [--placeholder]
                                [--install-argos] [--install-model]
python3 bookit_lang_tool.py strip-comments [FILE]
python3 bookit_lang_tool.py install-translator [--install-argos]

--root PATH    plugin root, defaults to the current directory
FILE          defaults to lang/en/bookit.php, and to lang/de/bookit.php
               for strip-comments
```

Functionality:

Audits, sorts, prunes and translates the language strings of mod_bookit. Put it in the plugin root and run it from there. Needs Python 3.10+ and the php binary, which it uses to tokenize our PHP so that commented-out calls are not counted as usages. Every command that writes creates a .bak next to the file first.

Notes:

unused prints one key per line, **report** prints counts only, and both take --json for the full result including a per-key list of every reference found. **sort** orders the assignments alphabetically and moves comments along with the string they sit above; --check reports without writing. **prune** deletes the unused keys and keeps the order of the rest, so the diff is deletions only. **sync** copies EN keys missing from DE and machine translates them with Argos, a local library rather than an API; --placeholder copies the English text instead, which is useful before Argos is set up. Every line sync writes carries a TODO marker that **strip-comments** removes later. Results go to stdout and warnings to stderr, so unused can be piped. Exit code is 1 when there is anything to act on.

What counts as a usage:

Literal references are read from get_string(), get_strings(), new lang_string(), moodle_exception() and print_error() in PHP; from getString(), get_strings() object arrays, prefetchStrings() and ternary calls in amd/src; and from {{#str}} and {{#cleanstr}} in templates. An addHelpButton() or help_icon() identifier also marks its _help and _link companions, but only if those exist, since both are optional. Capabilities from db/access.php, providers from db/messages.php and the standard keys count as used because Moodle reads them without a call site.

Dynamic keys are resolved from the call site outwards rather than assumed. The tool takes the actual expression and looks up its possible values in the code, covering 'resources:status_' . \$x, 'event_bookingstatus_' . \$x, the notification enum loop with the prefixes returned by get_notification_default_message_prefix(), DISPLAY_TABLE_COLUMNS, the ReflectionClass constant lookup in checklist_manager.php, and the customtemplatedefaultmessage_\${type} template literal in JS. Enum values, prefixes and status numbers come out of the source, so adding a case needs no change here. amd/build, thirdpartylibs, vendor and node_modules are skipped; scanning the build output would keep a string alive after its last real use was removed but before anyone ran grunt.

Limits:

Anything the tool cannot resolve is reported as a warning instead of being dropped silently, and **prune** refuses to run while such a warning is open, because the unresolved call site may be the only user of a key about to be deleted. Use --force once you have checked. One warning is expected right now: the reflection call in checklist_manager.php:229, whose target class has no constants at all.

What the tool does not do is follow data flow. A key assembled elsewhere and passed as get_string(\$key, ...), or hidden behind a wrapper, is flagged as unresolved rather than resolved. Before deleting a batch, look at rename_hazards in the JSON report: those keys have no literal reference anywhere, so grep will not find them either. Machine translations are drafts. Placeholders such as {\$a}, HTML tags and numbers are held out of the translation and put back afterwards, but the German still needs a read after using the tool.